```
clear; clc; close all;

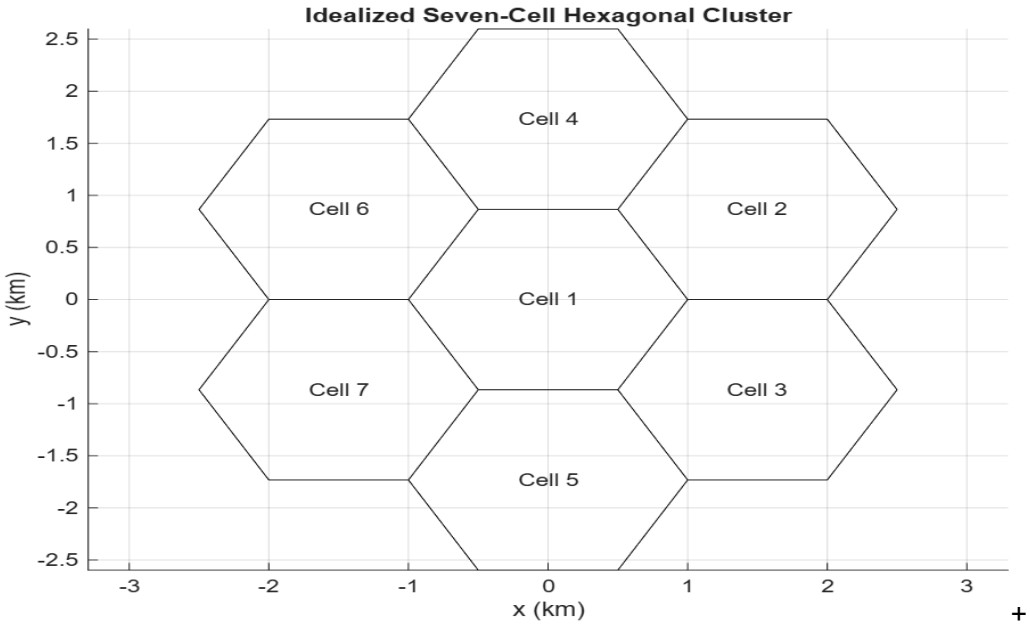
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
capacityTable = table(radiiKm(:), cellAreaKm2(:), cellCount(:), ...
    repmat(usersPerCell,numel(radiiKm),1), estimatedUsers(:), ...
    'VariableNames', {'Radius_km','CellArea_km2','EstimatedCells', ...
    'UsersPerCell','IllustrativeUsers'});
disp(capacityTable);

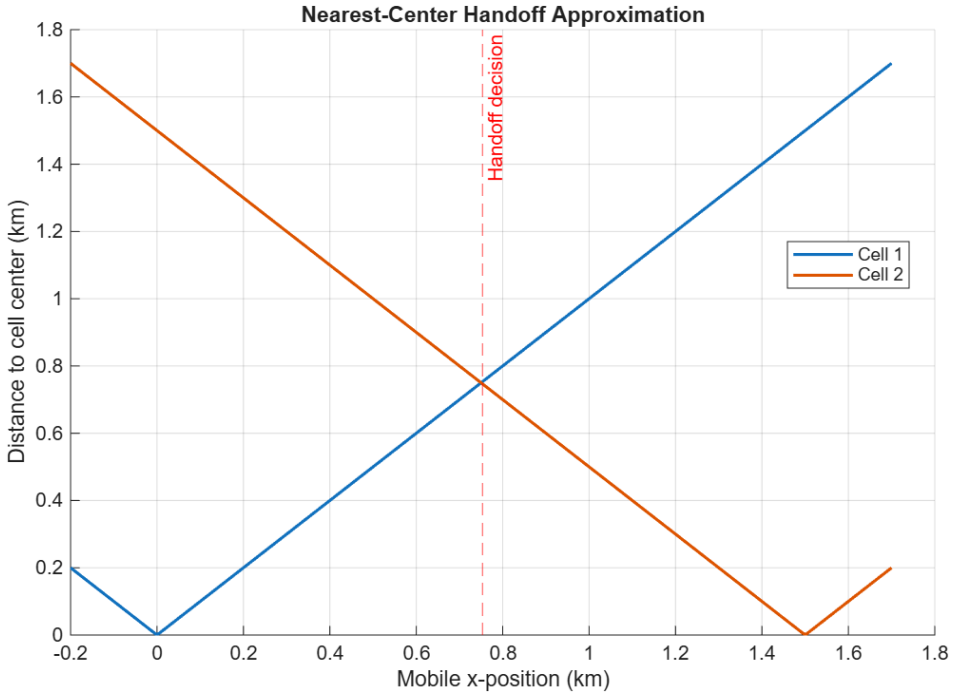
R = radiiKm(1);
centers = [0 0; 1.5*R sqrt(3)*R/2; 1.5*R -sqrt(3)*R/2; ...
    0 sqrt(3)*R; 0 -sqrt(3)*R; -1.5*R sqrt(3)*R/2; ...
    -1.5*R -sqrt(3)*R/2];
theta = (0:6)*pi/3;
figure; hold on; axis equal; grid on;
for k = 1:size(centers,1)
    plot(centers(k,1)+R*cos(theta), centers(k,2)+R*sin(theta), 'k-');
    text(centers(k,1),centers(k,2),sprintf('Cell %d',k), ...
        'HorizontalAlignment','center');
end
xlabel('x (km)'); ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');
hold off;

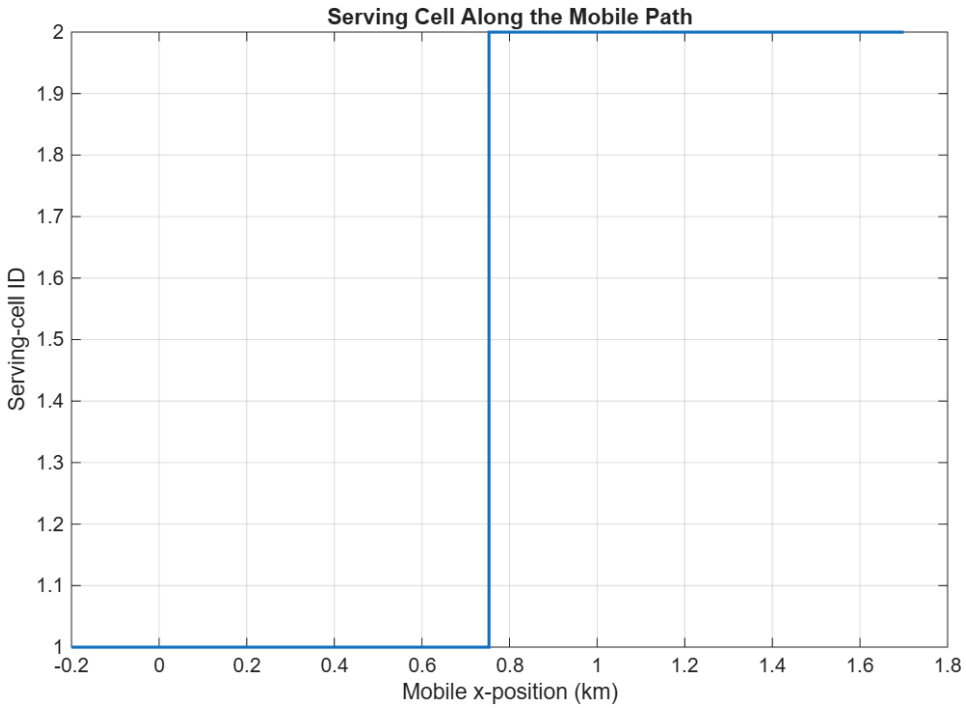
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX, mobileY);
d2 = hypot(mobileX-1.5*R, mobileY);
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
handoffIndex = find(diff(servingCell) ~= 0,1,'first') + 1;
figure; hold on; grid on;
plot(mobileX,d1,'LineWidth',1.5);
plot(mobileX,d2,'LineWidth',1.5);
if ~isempty(handoffIndex)
    xline(mobileX(handoffIndex),'--r','Handoff decision');
end
xlabel('Mobile x-position (km)'); ylabel('Distance to cell center (km)');
legend('Cell 1','Cell 2','Location','best');
title('Nearest-Center Handoff Approximation');
hold off;
figure; stairs(mobileX,servingCell,'LineWidth',1.5); grid on;
xlabel('Mobile x-position (km)'); ylabel('Serving-cell ID');
title('Serving Cell Along the Mobile Path');
if ~isempty(handoffIndex)
    fprintf('Geometric handoff at x = %.3f km\n',mobileX(handoffIndex));
end
```

Radius_km	CellArea_km2	EstimatedCells	UsersPerCell	IllustrativeUsers
1	2.5981	2	30	60
0.5	0.64952	8	30	240

```
Geometric handoff at x = 0.753 km
```




Nearest-Center Handoff Approximation
Handoff decision
Cell 1
Cell 2
Distance to cell center (km)
Mobile x-position (km)
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8


Serving Cell Along the Mobile Path
Serving-cell ID
Mobile x-position (km)
2
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1
-0.2
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8